NVIDIA H100 Tensor Core GPU - DATASHEET

Manufacturer: NVIDIA
Part Number: H100-SXM5
Category: GPU / Data Center GPU

Overview

The NVIDIA H100 is the flagship GPU of the Hopper architecture, delivering breakthrough performance for large-scale AI training and inference. It features 4th-generation Tensor Cores with FP8 precision and the Transformer Engine for automatic mixed-precision optimization. The H100 provides up to 9x faster AI training and 30x faster inference compared to A100. Status: ACTIVE - currently available for purchase.

Compute Specifications

CUDA Cores: 16,896. Tensor Cores: 528 (4th generation). Peak FP64: 34 TFLOPS. Peak FP32: 67 TFLOPS. Peak TF32 Tensor Core: 989 TFLOPS (with sparsity). Peak FP16 Tensor Core: 1,979 TFLOPS (with sparsity). Peak FP8: 3,958 TOPS (with sparsity). Transformer Engine: automatic FP8/FP16 mixed precision for transformer models.

Memory and Interconnect

HBM3 memory: 80 GB. Memory bandwidth: 3,350 GB/s (1.6x vs A100). Memory bus width: 5120-bit. L2 cache: 50 MB. NVLink 4.0: 18 links, 900 GB/s total bidirectional bandwidth. PCIe Gen5 x16 host interface. NVSwitch 3.0 support for 256 GPU clusters.

Multi-Instance GPU (MIG)

Second-generation MIG technology supports up to 7 GPU instances, each with dedicated compute, memory, and bandwidth resources. MIG instances are fully isolated at hardware level. Supported profiles: 1g.10gb, 2g.20gb, 3g.40gb, 4g.40gb, 7g.80gb. Each instance includes its own video decode engines.

Power, Thermal and Ordering

TDP: 700W (SXM5), 350W (PCIe). Max GPU temperature: 83 degrees C. Form factors: SXM5, PCIe. Price: $2,850 USD. Available from authorized NVIDIA partners and cloud service providers (AWS p5, Azure ND H100 v5, GCP A3). In stock: 45 units.